Case Study

Text Sentiment Classification Pipeline

Objective

Design and implement an end-to-end machine learning pipeline for text sentiment classification. Candidates may choose between open-source large language models (LLMs) or traditional machine learning approaches. The pipeline should be modular, scalable, and reproducible.

You are expected to explore the data, define the right questions, and guide the solution end-to-end, just like a consultant would. This is your opportunity to showcase not just your technical capabilities, but also your problem-solving mindset, communication skills, and business acumen.

This task is intentionally open-ended. You're encouraged to apply best practices, demonstrate initiative, and think critically about what matters.

Scope

- Data: Crypto News +
- Model Options (including but not limited to):
    - Open-source LLM (e.g., Hugging Face pipelines, Ollama, etc.) for sentiment scoring.
    - Traditional ML: Classical classifiers (e.g., logistic regression, SVM, Random Forest, RNN, etc.) with engineered features (TF-IDF, sentiment lexicons, etc.).
    - Hybrid Approach: Utilise LLM-generated embeddings and optionally combine them with traditional features, then feed into downstream classifiers.
- Languages & Environments: Python 3.8+
- Implementation: Python scripts or Jupyter notebooks
- Orchestration: MLflow, Prefect, Airflow, or custom scripts for ML pipeline.
- Hosting: Serve the selected model locally on localhost via an HTTP API (e.g., MLflow, FastAPI, etc.).
- Model Serving UI: Provide a web-based interface (e.g., Streamlit, Gradio, etc.) will be a plus.
- Visualisation: Power BI, Tableau, Streamlit, Gradio, Plotly or similar UI/BI frameworks.

The Task

You are required to:

- Design and implement an end-to-end ML pipeline for text sentiment classification, encompassing data ingestion, preprocessing, feature engineering, training/validation, and model serving (e.g. MLflow or similar tools).
- Ingest and preprocess data from the Kaggle crypto-news dataset, handling data cleansing, feature extraction, and class imbalance.
- Build and evaluate sentiment classifiers (positive, negative, neutral), optionally log experiments and metrics, and register the best model in a model registry.

- Develop an interactive dashboard to visualise sentiment trends over time, by cryptocurrency, and by source.

In addition, be prepared to present and walk us through your approach, including your assumptions, architectural decisions, and the rationale behind your insights - as you would in a client engagement.

What We're Looking For

This is not about perfection — we're more interested in **your approach**. We'll evaluate:

- **Architecture:** Is your end-to-end machine learning pipeline logical, modular, and scalable?
- **Model Selection & Evaluation:** Did you choose appropriate algorithms, justify your choices, and rigorously evaluate performance?
- **Data Cleansing & Feature Engineering:** How did you clean data, create, select, and manage features (e.g., text representations, embeddings, etc.)?
- **Assumptions & Communication:** Are your decisions well-reasoned and clearly explained in code, docs, and presentations?
- **Insight Delivery:** Is your dashboard interface clear, actionable, and story-driven?

No need to over-engineer, performance isn't the top priority if your choices are well justified.

Deliverables

There's no template for delivery, but your output might include:

- Provide a ZIP file or GitHub repository link containing the pipeline and model training scripts or notebooks.
- Include a README.md with a brief executive summary of the methodology, key findings, and recommendations.
- A dashboard.
- A showcase of your solution to our engineering team.

Good Luck

We look forward to seeing your creativity and technical expertise in action. Good luck!

If you have any questions about this case study, please contact subhashi.randeni@dataengine.co.nz. Understanding client expectations is a key skill for every consultant. 🙂